Business Recommendations Summary

Task 9 — Optimized Classification Model for Subscription-Likelihood Prediction

Business Problem

The company wants to identify which customers are most likely to become subscribed, repeat-engaged customers, so it can prioritize retention offers, personalize outreach, and avoid wasting marketing spend on unlikely candidates. An optimized Logistic Regression model was built and evaluated for this purpose (see the accompanying notebook and feature-importance report for full methodology and evidence).

Model Performance Supporting These Recommendations

Model	Status	Accuracy	Precision	Recall	F1-Score	ROC-AUC
Logistic Regression	Baseline	0.854	0.686	0.848	0.758	0.910
Decision Tree	Baseline	0.791	0.635	0.535	0.581	0.711
Random Forest	Baseline	0.844	0.663	0.858	0.748	0.907
Optimized Logistic Regression	Optimized	0.864	0.666	1.000	0.799	0.909
Optimized Random Forest	Optimized	0.864	0.666	1.000	0.799	0.909

The optimized Logistic Regression was selected as the final model: it matches the optimized Random Forest on F1-score and ROC-AUC while remaining fully interpretable.

Recommendation 1: Bundle Discounts Into Subscription Offers

Recommendation 2: Do Not Target on Gender — Validate First

Recommendation 3: Prioritize High-Likelihood Segments

Recommendation 4: Adopt the Optimized Pipeline for Scoring

Recommendation 5: Set the Threshold to Match Campaign Economics

Model finding	Discount Applied is the single strongest predictor of subscription status.
Business interpretation	Discounted customers are far more likely to be subscribed, consistent with subscription programs commonly including discount perks.
Recommended action	Bundle a modest discount into subscription sign-up offers, and test discount-led win-back offers for high-probability non-subscribers.
Expected benefit	Higher subscription conversion at a controlled, measurable promotional cost.
Risk / limitation	Discounts erode margin; roll out via an A/B test with a control group before a full rollout.

Model finding	Gender is the second strongest predictor, but every subscribed customer in this dataset is male.

Business interpretation	This is very likely a dataset-specific artifact, not a real driver of customer loyalty.
Recommended action	Do not use gender as a targeting criterion; re-validate this relationship against current, real transactional data before it informs any decision.
Expected benefit	Avoids biased, non-generalizable, and potentially unfair targeting decisions.
Risk / limitation	Acting on this pattern without validation risks both unfair treatment and poor real-world generalization.

Model finding	Customers segmented into a "High" subscription-likelihood band show a much higher actual subscription rate than the "Low" band.
Business interpretation	The model meaningfully separates loyalty-leaning customers from one-off shoppers.
Recommended action	Prioritize the High-likelihood segment for premium loyalty offers and early access; use lighter-touch, low-cost messaging for the Low segment.
Expected benefit	Better return on retention and marketing spend by matching offer intensity to likelihood.
Risk / limitation	Segment thresholds are a starting point and should be tuned against real campaign response data over time.

Model finding	Hyperparameter optimization (GridSearchCV, F1-scoring) improved F1-score over both baseline models.
Business interpretation	The tuned model identifies more true subscribers without a proportional rise in false positives.
Recommended action	Adopt the optimized Logistic Regression pipeline (documented hyperparameters, saved as purchase_prediction_model.pkl) as the production scoring model.
Expected benefit	More consistent, better-calibrated targeting decisions than the untuned baseline.
Risk / limitation	Hyperparameters were tuned on this single dataset extract and should be re-validated as new data arrives.

Model finding	Precision and recall trade off sharply as the classification threshold changes.
Business interpretation	The "right" cutoff depends on how expensive a wasted contact is versus the value of a missed loyal customer.
Recommended action	Set the operating threshold based on the real cost of a retention contact versus the value of retaining a subscriber, not the default 0.50.
Expected benefit	Retention spend is matched to the campaign's real economics rather than an arbitrary default.
Risk / limitation	Threshold choice needs input from marketing/finance on true costs, not model output alone.

Limitations

Subscription Status was adopted as a proxy target because the provided dataset has no literal purchase/no-purchase outcome or browsing-session data — these recommendations are about subscription likelihood, not raw purchase conversion. The dominant Gender effect appears to be a data-generation artifact rather than a real behavioural pattern and must be re-validated on live data before informing any targeting decision. All results come from a single static extract of 3,900 customers; performance should be monitored and the model retrained as fresh data becomes available. Feature importance reflects association, not proven causation.